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Instructions:

Read Lesson 2: Quantitative Research Designs. Then, answer the following tasks. Provide detailed explanations and examples to justify your answers.

Part A: Analysis & Application

1. **Compare and Contrast** – Differentiate correlational and causal-comparative designs in terms of purpose, control of variables, and limitations. Provide an example study for each in the field of education or IT.

Compare and Contrast Correlational design is used to see if two variables are related, but it does not show cause and effect. Causal-comparative design compares groups after an event has happened and also cannot control variables. The difference is that correlation focuses on the strength of relationships, while causal-comparative focuses on differences between groups. Example of correlational study hours and grades. Example of causal-comparative performance of public and private school students.

2. **Design Critique** – Suppose a researcher uses a descriptive design to test the effectiveness of a new teaching strategy. Explain why this design may be inappropriate. Suggest a better design and justify your choice.

Judgment Task Survey design is useful, but it is not the most powerful. It cannot prove cause and effect and depends on honest answers. Other designs, like experimental, can give stronger evidence of effectiveness.

Part B: Evaluation

1. **Case Scenario** – DNSC wants to study whether the number of study hours affects exam performance among first-year students.

Case Scenario The appropriate design is correlational because it studies the relationship between study hours and exam performance. The independent variable is study hours. The dependent variable is exam scores. An extraneous variable could be sleep, because it also affects performance.

2. **Judgment Task** – A researcher claims that survey design is the most powerful quantitative design because it can be applied to any situation. Do you agree or disagree? Provide at least two reasons supported by examples.

Judgment Task Survey design is useful, but it is not the most powerful. It cannot prove cause and effect and depends on honest answers. Other designs, like experimental, can give stronger evidence of effectiveness.

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Part C: Creation – Mini Research Proposal (Outline Only)

1. Title of the Study

The Role of ICT in Enhancing User Competence among BSIT 3rd Year Students at Davao del Norte State College

2. Research Question(s)

1. What is the level of ICT use among BSIT 3rd year students?
2. What is the level of user competence among BSIT 3rd year students?
3. Is there a significant relationship between ICT use and user competence?

3. Design Chosen and Justification

Design: Descriptive-Correlational Survey

Justification:

- A **descriptive approach** will describe the current level of ICT use and user competence of the respondents.
- The **correlational aspect** will determine if there is a significant relationship between ICT use (independent variable) and user competence (dependent variable), which fits the research purpose without manipulating variables.

4. Variables

- **Independent Variable (IV):** ICT Use/Role
  - Indicators: Access & availability, frequency of use, diversity of tools, quality of integration
- **Dependent Variable (DV):** User Competence
  - Indicators: Technical skills, problem-solving, ICT literacy, self-confidence, productivity
- **Control Variables (if applicable):** Year level (limited to 3rd-year BSIT students), program (BSIT), same academic year.